

BST Warmups 02

Empty BST:

Insert 9:

⑨

2 inserts left of 9 because its smaller

Insert 2:

② ⑨

Will continue inserting left

Insert 1:

① ② ⑨

10 will insert right of 9 because its larger

Insert 10:

① ② ⑨ ⑩

8 is smaller than 9 but larger than 2 so it goes left then right

Insert 8:

① ② ⑧ ⑨ ⑩

3 is smaller than 9 larger than 2 so it goes left, right smaller than 8 so it goes left

Insert 3:

① ② ⑧ ⑨ ⑩
③

Insert 7:

① ② ⑧ ⑨ ⑩
③ ⑦

7 will follow the trend and insert right of 5 because its larger

Insert 4:

① ② ⑧ ⑨ ⑩
③ ④ ⑦

4 will follow the trend and finally insert left of 7 because its smaller

Insert 6:

① ② ⑧ ⑨ ⑩
③ ④ ⑦ ⑥

6 will follow the trend and insert right of 4 because its larger

Insert 5:

① ② ⑧ ⑨ ⑩
③ ④ ⑦ ⑤ ⑥

Will insert left of 6 due to it being smaller

Delete 4:

① ② ⑧ ⑨ ⑩
③ ⑦ ⑤ ⑥

Since 10 is the rightmost node it will become the root

Insert 9:

① ② ⑩ ⑧ ⑨ ⑦ ⑤ ⑥

Delete Root:

① ② ⑧ ⑨ ⑦ ⑤ ⑥

2 will become the only root because its the only branch of 10

Delete Root Again:

③ ① ⑧ ⑨ ⑦ ⑤ ⑥

3 becomes of the general rule

- 1. Go rightmost node
- 2. Go as far left as possible

the smallest value of the subtree is 3 and 7 moves up to replace 3